MPG No. 13.5.130 $4'm'm$ (4'mm' setting) [Type III, tetragonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & -T_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{xx} = \mathbb{T}_{v}^{(1)}$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ 0 & -T_{xxz} & 0 \\ T_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{xxz} = \mathbb{T}_{z}^{\beta(1)}$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ 0 & \frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{xxz} = 2\mathbb{M}_{xy}^{(1)}$

$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)}$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ T_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{yzy} = \mathbb{M}_{xy}^{(2)}$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & 0 & T_{xxzz} & 0 & 0 & 0 \\ 0 & -T_{xxxx} & -T_{xxzz} & 0 & 0 & 0 \\ T_{xxzz} & -T_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$T_{xxxx} = -\mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_{v}^{(1)} + 4\mathbb{T}_{v}^{(2)}$

$T_{xxzz} = \mathbb{T}_{4v}^{(1)} + \mathbb{T}_{v}^{(1)}$

$T_{yzyz} = -\mathbb{T}_{4v}^{(1)} - \mathbb{T}_{v}^{(2)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 \\ -T_{xxyy} & 0 & -T_{xxzz} & 0 & 0 & 0 \\ -T_{xxzz} & T_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxyy} = 4\mathbb{M}^{(1)}_{xyz} - 2\mathbb{T}^{(3)}_{v}$$
$$T_{xxzz} = -4\mathbb{M}^{(1)}_{xyz} - \mathbb{T}^{(3)}_{v}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & 0 \\ 0 & -T_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzyz} = 2\mathbb{T}^{(4)}_{v}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzyz} & 0 & 0 \\ 0 & T_{yzyz} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}^{(2)}_{xyz} - \mathbb{T}^{(5)}_{v}$$
$$T_{xyxy} = \mathbb{M}^{(2)}_{xyz} + 2\mathbb{T}^{(5)}_{v}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(6)}_{v}$$
$$T_{xyxy} = \mathbb{M}^{(3)}_{xyz} + 2\mathbb{T}^{(6)}_{v}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxyy} & -T_{xxxx} & -T_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{zzxx} & -T_{zzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{yzzy} & 0 & 0 & -T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & -T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = -\mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{v} + 4\mathbb{T}^{(2)}_{v}$$
$$T_{xxyy} = 4\mathbb{M}^{(1)}_{xyz} - 2\mathbb{T}^{(3)}_{v}$$
$$T_{xxzz} = -4\mathbb{M}^{(1)}_{xyz} + \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(1)}_{v} - \mathbb{T}^{(3)}_{v}$$
$$T_{zzxx} = 4\mathbb{M}^{(1)}_{xyz} + \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v}$$
$$T_{yzyz} = \mathbb{M}^{(2)}_{xyz} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(2)}_{v} - \mathbb{T}^{(5)}_{v}$$
$$T_{yzzy} = -\mathbb{M}^{(2)}_{xyz} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v}$$
$$T_{xyxy} = \mathbb{M}^{(2)}_{xyz} + 2\mathbb{T}^{(5)}_{v}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzzy} & 0 & 0 & T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}^{(3)}_{xyz} + 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v}$$
$$T_{yzzy} = \mathbb{M}^{(3)}_{xyz} - 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v}$$
$$T_{xyxy} = \mathbb{M}^{(3)}_{xyz} + 2\mathbb{T}^{(6)}_{v}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & 0 & 0 \\ 0 & -T_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{yyzz} \\ T_{zzxx} & 0 & 0 \\ 0 & -T_{yyzz} - T_{zzxx} & 0 \\ 0 & T_{yzzy} & 0 \\ 0 & 0 & T_{zxxz} \\ -T_{yzzy} - T_{zxxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = -\mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{v} + 4\mathbb{T}^{(2)}_{v}$$
$$T_{yyzz} = -\mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v}$$
$$T_{zzxx} = \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v}$$
$$T_{yzzy} = -\mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v}$$
$$T_{zxxz} = \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v}$$